I421 Applications in Data Mining

Homework 5

Total points: 50

**Note: this assignment needs to use R script, please write your answer on the R script file with enough comments. If you need to use word file to summary your answer, you need to zip your word file and R script together and submit on Canvas.**

Question 1 (10 points)

**(a)** Write functions `tmpFn1` and `tmpFn2` such that if `xVec` is the vector $(x_1, x_2, \ldots, x_n)$, then `tmpFn1(xVec)` returns the vector $(x_1, x_2^2, \ldots, x_n^n)$ and `tmpFn2(xVec)` returns the vector $\left(x_1, \frac{x_2^2}{2}, \ldots, \frac{x_n^n}{n}\right)$.

**(b)** Now write a function `tmpFn3` which takes 2 arguments `x` and `n` where `x` is a single number and `n` is a strictly positive integer. The function should return the value of

$$1 + \frac{x}{1} + \frac{x^2}{2} + \frac{x^3}{3} + \cdots + \frac{x^n}{n}$$

Question 2 (10 points)

Consider the continuous function

$$f(x) = \begin{cases} x^2 + 2x + 3 & \text{if } x < 0 \\ x + 3 & \text{if } 0 \le x < 2 \\ x^2 + 4x - 7 & \text{if } 2 \le x. \end{cases}$$

Write a function `tmpFn` which takes a single argument `xVec`. The function should return the vector of values of the function $f(x)$ evaluated at the values in `xVec`.
Hence plot the function $f(x)$ for $-3 < x < 3$.

Question 3 (10 points)

Write a function which takes a single argument which is a matrix. The function should return a matrix which is the same as the function argument but every odd number is doubled.

Hence the result of using the function on the matrix

$$\begin{bmatrix} 1 & 1 & 3 \\ 5 & 2 & 6 \\ -2 & -1 & -3 \end{bmatrix}$$

should be:

$$\begin{bmatrix} 2 & 2 & 6 \\ 10 & 2 & 6 \\ -2 & -2 & -6 \end{bmatrix}$$

*Hint:* First try this for a specific matrix on the Command Line.

Question 4 (20 points)

*Time Series.* The code

```
ts(datVec, start=c(1960,3), frequency=12)
```

creates a time series with monthly observations (`frequency=12`), with first observation in March 1960 (`start=c(1960,3)`) and with values specified in the vector `datVec`.

Suppose $z_1, z_2, \ldots, z_n$ is a time series. Then we define the exponentially weighted moving average of this time series as follows: select a starting value $m_0$ and select a discount factor $\delta$. Then calculate $m_1, m_2, \ldots, m_n$ recursively as follows: for $t = 1, 2, \ldots, n$

$$e_t = z_t - m_{t-1}$$
$$m_t = m_{t-1} + (1 - \delta)e_t$$

**(a)** Write a function `tsEwma(tsDat, m0=0, delta=0.7)` where `tsDat` is a time series, `m0` is the starting value $m_0$ and `delta` is $\delta$. The function should return $m_1, m_2, \ldots, m_n$ in the form of a time series.

**(b)** In general, *looping over named objects is much slower than looping over objects which do not have names.* This principle also applies to time series: looping over a vector is much quicker than looping over a time series. Use this observation to improve the execution speed of your function which should still return a time series. Investigate the difference in speed between the functions in parts (a) and (b) by using the function `system.time`.